UNIVERSITY OF RWANDA

COLLEGE OF SCIENCE AND TECHNOLOGY

COMPUTER ENGINEERING

MOBILE APPICATIONS AND DESIGN

LAB 6

REGO:223006785

Observations

During the development and testing of push notifications using Firebase cloud Messaging in flutter application, I made the following observations.

(1) Device token

during I observed that each device generates a unique fcm token. This token is very important because it is used to send notifications directly to a specific value. I also noticed that the token can change when the application is reinstalled or when the app data is cleared.

(2) The behaviour of notifications depends on the app state

- When the app is open, the notification does not appear automatically, so I display it using a popup.

- When the app is in the background or closed, the notification appears in the system notification bar.

(3) Notifications worked correctly on a real android device, but not on chrome, which showed that fcm is mainly for mobile apps.

(4) An internet connection and notification permission are required for the app to receive messages.

(5) When I sent messages from the firebase console, they were received instantly if the correct token used.

- using logs helped me confirm token and messages were working correctly.

Key steps followed.

1. Create flutter application
2. Create Firebase project
   - Open Firebase console
   - Click Add project
   - Enter project name
   - Click create project

3. Register Android App
   - Click Android Icon in firebase
   - Enter package name.
   - Download google-services.json
   - Place it in android/app/

(4) Install firebase CLI

- I already have a node JS
- I installed firebase-tools

(5) login and Access projects

by firebase login

(6) Configure flutter with firebase

dart pub global activate flutterfire_cli

Then

flutterfire configure

which

- Connected app to firebase
- Generates firebase_options.dart
- Configures Android automatically.

(7) Add dependencies of

- Firebase_core
- firebase_messaging.

(8) Initialize firebase in main.dart

(9) Request notification permission

10. Retrive fcm Token

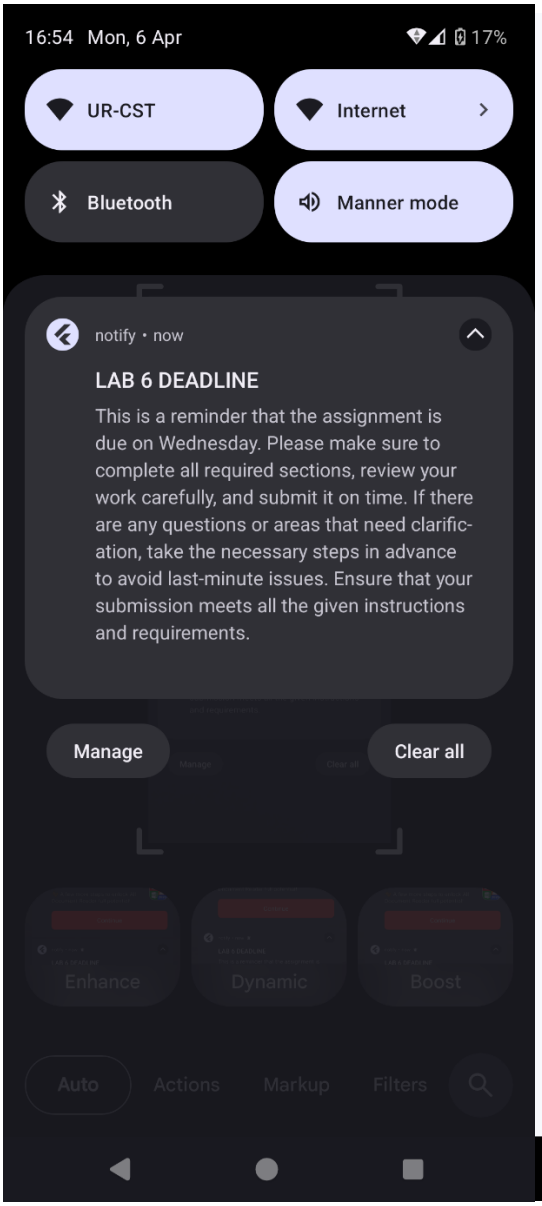
11. Hand Notifications

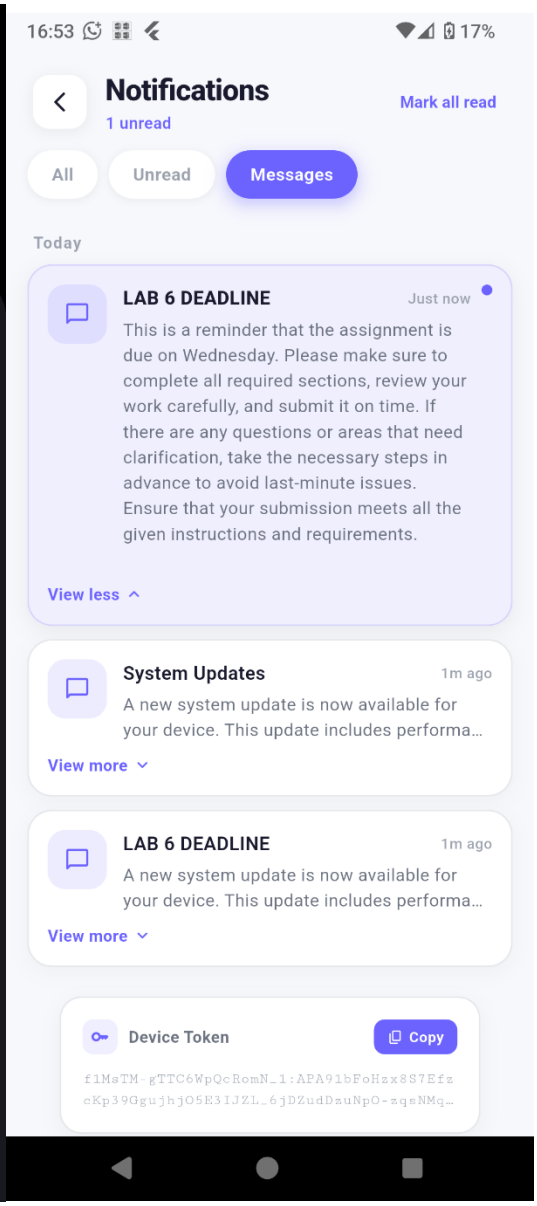
12. Display Notification

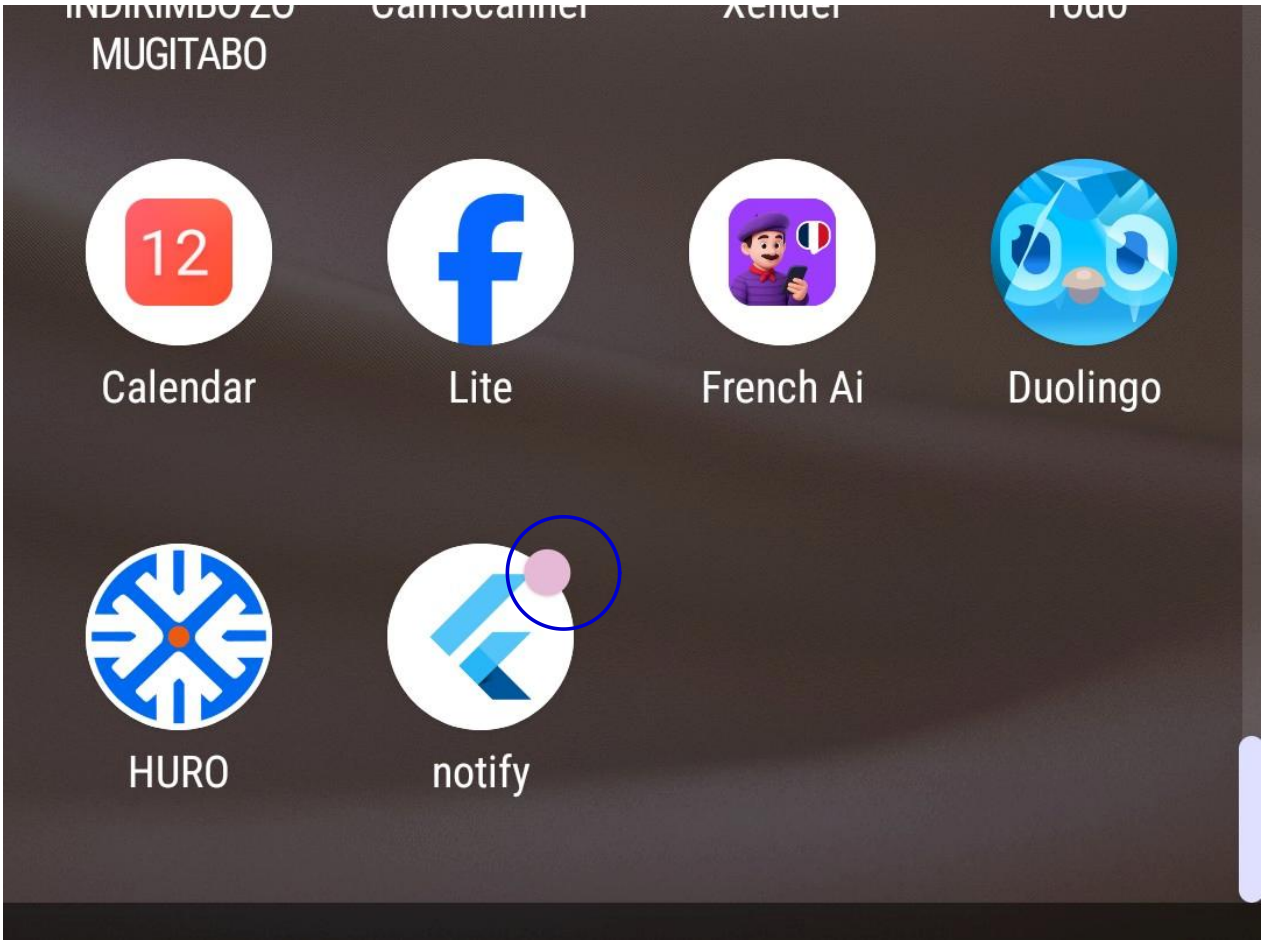
13. Send test Notification.

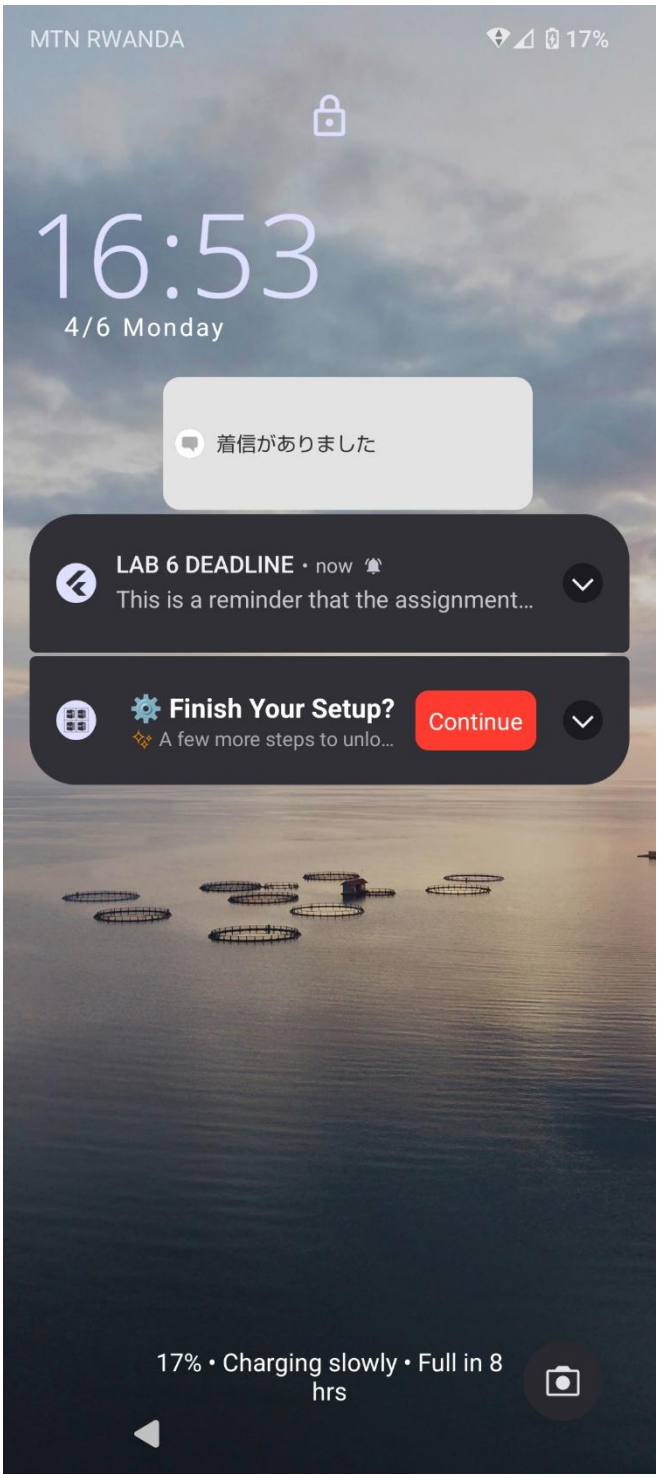
- Firebase console
- Open Cloud messaging
- Create a message
- Add device token
- send.

Screenshots of the application

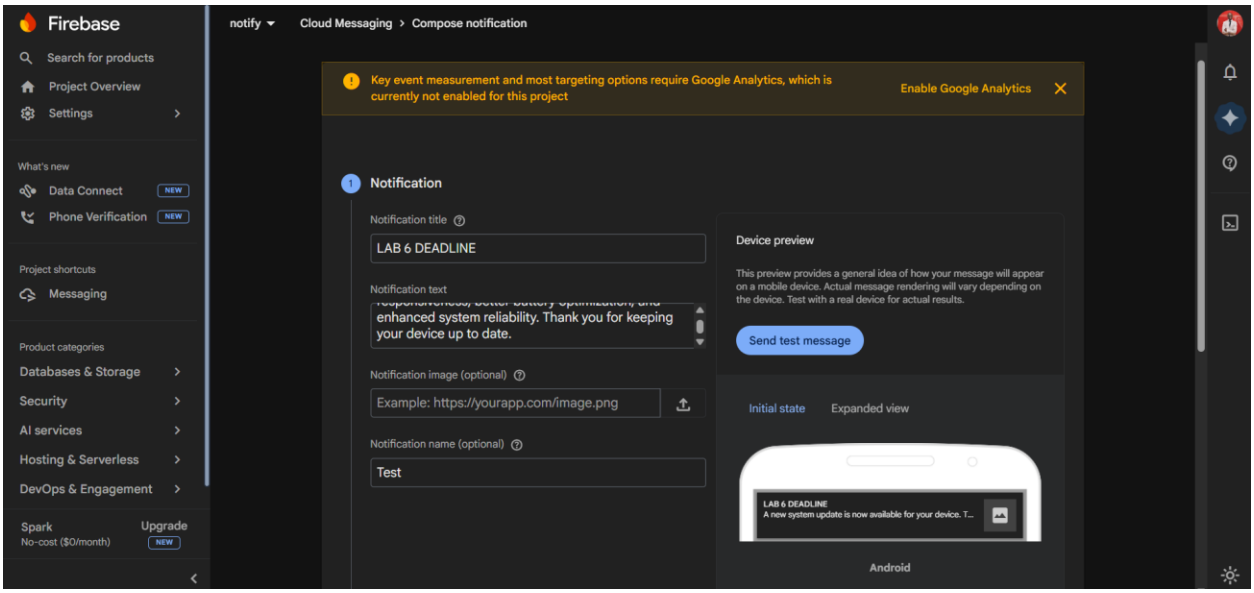

16:54 Mon, 6 Apr
17%
UR-CST
Internet
Bluetooth
Manner mode
notify • now
LAB 6 DEADLINE
This is a reminder that the assignment is due on Wednesday. Please make sure to complete all required sections, review your work carefully, and submit it on time. If there are any questions or areas that need clarification, take the necessary steps in advance to avoid last-minute issues. Ensure that your submission meets all the given instructions and requirements.
Manage
Clear all
Enhance
Dynamic
Boost
Auto
Actions
Markup
Filters

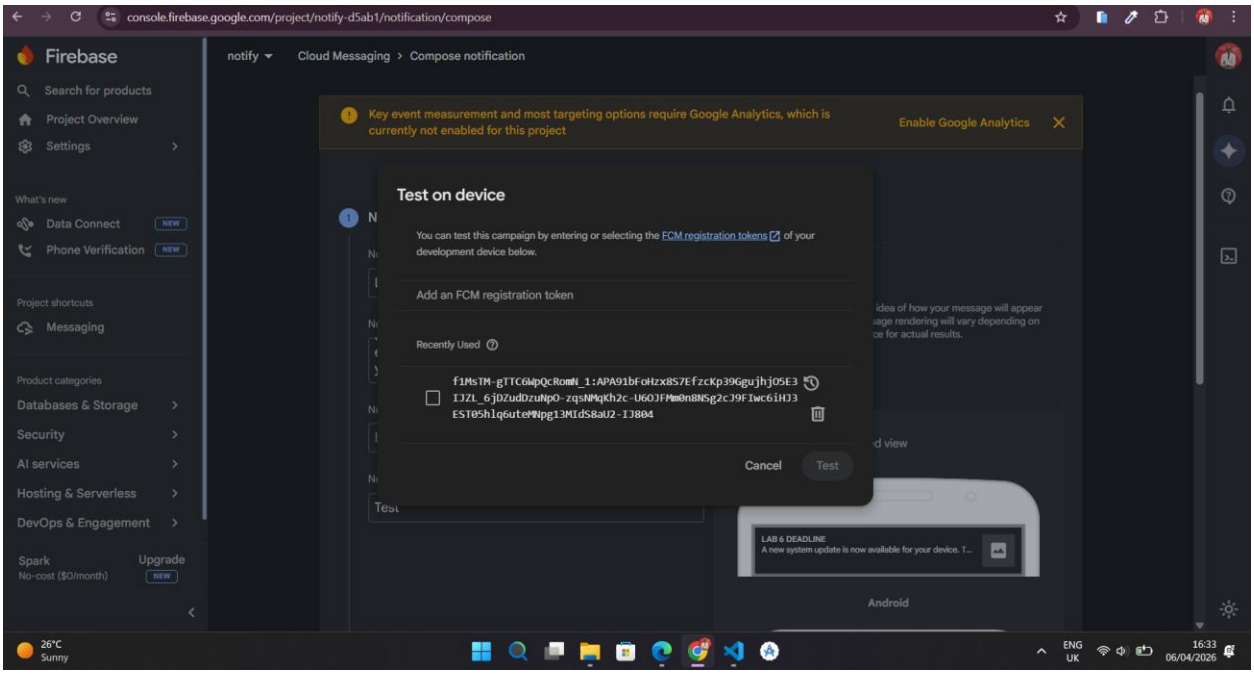

16:53
17%
Notifications
1 unread
Mark all read
All
Unread
Messages
Today
LAB 6 DEADLINE
Just now
This is a reminder that the assignment is due on Wednesday. Please make sure to complete all required sections, review your work carefully, and submit it on time. If there are any questions or areas that need clarification, take the necessary steps in advance to avoid last-minute issues. Ensure that your submission meets all the given instructions and requirements.
View less
System Updates
1m ago
A new system update is now available for your device. This update includes performa...
View more
LAB 6 DEADLINE
1m ago
A new system update is now available for your device. This update includes performa...
View more
Device Token
Copy
f1MsTM-gTTC6WpQcRomN_1:APA91bFoHzx8S7Efz
cKp39GgujhjO5E3IJZL_6jDZudDzuNpO-zqsNMq...


MUGITABO
Calendar
Lite
French Ai
Duolingo
HURO
notify



Screenshots of Firebase Cloud Messaging interface


Firebase
notify
Cloud Messaging > Compose notification
Search for products
Project Overview
Settings
What's new
Data Connect NEW
Phone Verification NEW
Project shortcuts
Messaging
Product categories
Databases & Storage
Security
AI services
Hosting & Serverless
DevOps & Engagement
Spark
No-cost ($0/month)
Upgrade NEW
Key event measurement and most targeting options require Google Analytics, which is currently not enabled for this project
Enable Google Analytics
Notification
Notification title
LAB 6 DEADLINE
Notification text
enhanced system reliability. Thank you for keeping your device up to date.
Notification image (optional)
Example: https://yourapp.com/image.png
Notification name (optional)
Test
Device preview
This preview provides a general idea of how your message will appear on a mobile device. Actual message rendering will vary depending on the device. Test with a real device for actual results.
Send test message
Initial state
Expanded view
LAB 6 DEADLINE
A new system update is now available for your device. T...
Android


console.firebase.google.com/project/notify-d5ab1/notification/compose
Firebase
notify
Cloud Messaging > Compose notification
Key event measurement and most targeting options require Google Analytics, which is currently not enabled for this project
Enable Google Analytics
Test on device
You can test this campaign by entering or selecting the FCM registration tokens of your development device below.
Add an FCM registration token
Recently Used
f1MsTM-gTTC6WpQcRomN_1:APA91bFoHzx8S7EfzcKp39GgujhjO5E3
IJZL_6jDZudDzuNpO-zqsNMqKh2c-U6OJFMm0n8N5g2cJ9FIwc6iHJ3
ESTO5hlq6uteMNpg13MIdS8aU2-IJ804
Cancel
Test
LAB 6 DEADLINE
A new system update is now available for your device. T...
Android
26°C
Sunny
ENG
UK
16:33
06/04/2026